F18CE2 - MV Calculus and Real Analysis B: Solutions 2

1 Let $\epsilon > 0$. Since $\lim_{n\to\infty} b_n = 0$, there exists N such that, for $n \geqslant N$, $|b_n - 0| < \dfrac{\epsilon}{K}$, and so

$$|a_n| \leqslant K|b_n| < \epsilon.$$

2 Let $\epsilon > 0$. Now,

$$|a_n + b_n - L - M| = |(a_n - L) + (b_n - M)| \leqslant |a_n - L| + |b_n - M|.$$

Since $\lim_{n\to\infty} a_n = L$ and $\lim_{n\to\infty} b_n = M$, there exist $N_a,\ N_b$ such that

$$|a_n - L| < \frac{\epsilon}{2} \quad \text{for } n \geqslant N_a \quad \text{and} \quad |b_n - M| < \frac{\epsilon}{2} \quad \text{for } n \geqslant N_b.$$

Let $N = \max\{N_a,\ N_b\}$. Then for $n \geqslant N$,

$$|a_n + b_n - L - M| \leqslant |a_n - L| + |b_n - M| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

3 We first note that

$$|a_n| = |a_n - L + L| \leqslant |a_n - L| + |L|,$$

so we get what we want if we can make $|a_n - L| \leqslant 1$. To do this we set $\epsilon = 1$ in the definition of the limit (this definition works for any $\epsilon > 0$, so we are allowed to do this). Then, by definition, there is an N such that, for $n \geqslant N$, $|a_n - L| < 1$ and hence, by the above calculation, $|a_n| < 1 + |L|$.

4 Let $\epsilon = L/2 > 0$ in the definition of limit. Then there exists an integer N such that

$$|a_n - L| < \epsilon = \frac{L}{2} \quad \text{for all } n \geqslant N.$$

In particular, for $n \geqslant N$,

$$-\frac{L}{2} < a_n - L, \quad \text{so} \quad \frac{L}{2} < a_n.$$

For the second part let $\epsilon > 0$. From above, if $n \geqslant N$ then

$$\left|\frac{1}{a_n} - \frac{1}{L}\right| = \frac{|a_n - L|}{a_n L} < \frac{2|a_n - L|}{L^2}.$$

From the definition of a limit, there exists an integer N_1 such that $|a_n - L| < \epsilon L^2/2$ for $n \geqslant N_1$.

Hence if $n \geqslant N_2 = \max\{N, N_1\}$, then

$$\left|\frac{1}{a_n} - \frac{1}{L}\right| < \frac{2|a_n - L|}{L^2} < \epsilon.$$

5 A is bounded above if there exists K_u such that $a \leqslant K_u$ for all $a \in A$. $\sup A$ is the least K_u.

6 (a) False, the set $\mathbb{R}$ does not have a supremum.

(b) True, $a \leqslant M$ for all $a \in A$ so $a \leqslant L$ for all $a \in A$.

7 No, for example take $A = \{-1, 0, 3\},\ B = \{-1, 3\}$.

8 We prove that $\sup A = 2$.

For $a \in A$, we have $a < 2$ so that 2 is an upper bound for A.

Now let $\epsilon > 0$. We must show that $2 - \epsilon$ is not an upper bound for A.

The number $2 - \epsilon/2$ is in A and $2 - \epsilon/2 > 2 - \epsilon$.

Hence, $2 - \epsilon$ is not an upper bound for A so $\sup A = 2$.

9 We do the usual two steps to show that a number is a sup.

(i) We have that $a \leqslant \sup A$ for any $a \in A$, so if $b \in B$ then

$$b = 3a + 1 \leqslant 3 \sup A + 1.$$

so $3 \sup A + 1$ is an upper bound for B.

(ii) Now let $\epsilon > 0$. By definition, there is a number $a \in A$ such that $a > \sup A - \epsilon$.

Then, $3a + 1 \in B$ and $3a + 1 > 3 \sup A + 1 - \epsilon$, so $3 \sup A + 1 - \epsilon$ is not an upper bound for B.

Combining (i) and (ii) shows that $\sup B = 3 \sup A + 1$, by Remarks 1.40 in the notes.

Alternatively:

(iii) Choose a sequence (a_n) in A such that $a_n \to \sup A$. Then the sequence $(3a_n + 1)$ is in B, and $3a_n + 1 \to 3 \sup A + 1$.

Combining (i) and (iii) shows that $\sup B = 3 \sup A + 1$, by Theorem 1.43 in the notes.

10 Again, we do the usual two steps to show that a number is an inf. We will only use the method from Theorem 1.43 in the notes for this example.

(i) We have that $0 < a \leqslant \sup A$ for any $a \in A$, so if $b \in B$ then

$$b = \frac{1}{1 + a} \geqslant \frac{1}{1 + \sup A},$$

so $1/(1 + \sup A)$ is a lower bound for B.

(ii) Choose a sequence (a_n) in A such that $a_n \to \sup A$, and let $b_n = 1/(1 + a_n)$, for $n \geqslant 1$. Then the sequence (b_n) is in B, and $b_n = 1/(1 + a_n) \to 1/(1 + \sup A)$.

Combining (i) and (ii) shows that $\inf B = 1/(1 + \sup A)$.